

Assignment – 6

Submitted by ~ Ishika Dutta (CT_CSI_DV_5572)

Task 6.1

Deploy Replica Set and Replication Controller, and deployment. Also learn the advantages and disadvantages of each.

Replication Controller

```
1 kubectl apply -f rc.yaml
2
```

Example Use Case:

```
1  apiVersion: v1
2  kind: ReplicationController
3  metadata:
4    name: my-rc
5  spec:
6    replicas: 3
7    selector:
8      app: nginx
9    template:
10     metadata:
11       labels:
12         app: nginx
13     spec:
14       containers:
15         - name: nginx
16           image: nginx
17
```

Replica Set

```
1  apiVersion: apps/v1
2  kind: ReplicaSet
3  metadata:
4    name: my-replicaset
5  spec:
6    replicas: 3
7    selector:
8      matchLabels:
9        app: nginx
10    template:
11     metadata:
12       labels:
13         app: nginx
14     spec:
15       containers:
16         - name: nginx
17           image: nginx
18
```

Deployment

```

1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: nginx-deployment
5  spec:
6    replicas: 3
7    selector:
8      matchLabels:
9        app: nginx
10   template:
11     metadata:
12       labels:
13         app: nginx
14     spec:
15       containers:
16       - name: nginx
17         image: nginx:latest
18

```

Task 6.2

Kubernetes service types (ClusterIP, NodePort, LoadBalancer)

ClusterIP

```

spec:
  type: ClusterIP

```

NodePort

```

spec:
  type: NodePort
  ports:
    - port: 80
      nodePort: 30080

```

LoadBalancer

```

spec:
  type: LoadBalancer

```

Task 6.3

Kubernetes service types (ClusterIP, NodePort, LoadBalancer)

(Repetition, Same as Task 6.3)

Task 6.4

Managing Kubernetes with Azure Kubernetes Service (AKS), Creating and managing AKS clusters, Scaling and upgrading AKS clusters

Creating

```
az aks create \  
  --resource-group myResourceGroup \  
  --name myAKSCluster \  
  --node-count 3 \  
  --enable-addons monitoring \  
  --generate-ssh-keys
```

Managing

```
az aks get-credentials --name myAKSCluster --resource-group myResourceGroup
```

Scaling

```
az aks scale --resource-group myResourceGroup --name myAKSCluster --node-count 5
```

Updating

```
az aks upgrade --resource-group myResourceGroup --name myAKSCluster --kubernetes-version <version>
```

Task 6.5

Configure liveness and readiness probes for pods in AKS cluster.

```
1  <!-- livenessProbe -->  
2  <!-- httpGet -->  
3    path: /healthz  
4    port: 8080  
5    initialDelaySeconds: 5  
6    periodSeconds: 10  
7  
8  <!-- readinessProbe -->  
9  <!-- httpGet -->  
10   path: /ready  
11   port: 8080  
12   initialDelaySeconds: 5  
13   periodSeconds: 10  
14
```

Task 6.6

Configure Taints and Tolerants.

Taints

```
kubectl taint nodes node1 key=value:NoSchedule
```

Tolerants

```
1  ✓ tolerations:
2  ✓ - key: "key"
3    operator: "Equal"
4    value: "value"
5    effect: "NoSchedule"
6
```

Task 6.7

Create and attach persistent volume claims to pods.

Create Persistent Volume

```
  apiVersion: v1
  kind: PersistentVolume
  ✓ metadata:
    name: my-pv
  ✓ spec:
  ✓ capacity:
    storage: 1Gi
  ✓ accessModes:
    - ReadWriteOnce
  ✓ hostPath:
    path: /mnt/data
```

Create PersistentVolumeClaim

```
1  apiVersion: v1
2  kind: PersistentVolumeClaim
3  ✓ metadata:
4    name: my-pvc
5  ✓ spec:
6  ✓ accessModes:
7    - ReadWriteOnce
8  ✓ resources:
9  ✓ requests:
10    storage: 1Gi
11
```

Attach to Pod

```
1 volumes:
2   - name: data
3     persistentVolumeClaim:
4       claimName: my-pvc
5
```

Task 6.8

Configure health probes for pods.

```
1 livenessProbe:
2   exec:
3     command:
4     - cat
5     - /tmp/healthy
```

Task 6.9

Configure autoscaling in your cluster (Horizontal scaling)

Enable Metrics Server

Install metrics-server in your cluster (for AKS, it's available by default sometimes).

Create HPA

```
kubectl autoscale deployment nginx-deployment --cpu-percent=50 --min=1 --max=10
```

View HPA

```
kubectl get hpa
```